

2(a)(i)	1 <i>similarity:</i> both involve a (signalling), chemical / ligand / molecule ; 2 <i>difference:</i> endocrine chemical / hormone, travels in <u>blood / circulatory system</u> (but paracrine chemical travels by diffusion) ;	2
2(a)(ii)	1 (ACh) travels a short distance by <u>diffusion</u> ; 2 across, synapse / synaptic cleft / neuromuscular junction ;	2
2(b)	<i>any three conclusions from:</i> 1 palmitate-exposed cells, release / secrete, molecule / chemical / substance ; 2 (this) molecule switches on stress genes in, new / other, muscle cells ; 3 molecule is not, destroyed / denatured, by, boiling / high temperature ; 4 (so) molecule is not a protein ; 5 molecule is, lipid / lipid-soluble / hydrophobic / non-polar ; 6 AVP ;	3
2(c)	<i>any three from:</i> 1 less actin ; 2 less myosin ; 3 less troponin ; 4 less tropomyosin ; 5 fewer / smaller, sarcomeres / myofibrils / myofilaments / mitochondria / ribosomes ; 6 less of other named protein ;	3

Question	Answer	Marks
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